

Retail Revenue & Profit Intelligence

Complete Explanation of the 10 SQL Queries

PROJECT OVERVIEW

Database: retail_revenue_profit **Table:** superstore

This project uses SQL to analyze retail transaction data and convert raw records into business insights about sales, profit, products, categories, regions, customer segments, monthly trends, discounts, high-value customers and business growth.

BUSINESS PROBLEMS SOLVED

- Where is revenue and profit coming from?
- Which products and categories are the strongest?
- Which regions and customer segments perform best?
- Which products are causing losses?
- Is discounting affecting profitability?
- Which customers are high value?
- Is the business growing or declining?

DATA CLEANING

Before analysis, the project checks record count, NULL values, duplicate Row IDs, blank product/customer names, invalid discounts, categories and the order-date range. This improves data quality before business analysis.

EXPLANATION OF ALL 10 QUERIES

01. OVERALL BUSINESS PERFORMANCE

Business Question: What are the overall sales, profit, orders, and quantity sold by the business?

What we used: SUM(Sales), SUM(Profit), COUNT(DISTINCT Order ID), SUM(Quantity), ROUND().

What the query does: Calculates the main business KPIs and gives a high-level snapshot of business performance.

02. CATEGORY PERFORMANCE

Business Question: Which product categories generate the highest sales and profit?

What we used: GROUP BY Category, SUM(Sales), SUM(Profit), ORDER BY.

What the query does: Groups categories and compares their revenue and profitability.

03. TOP-SELLING PRODUCTS

Business Question: Which are the top 10 products contributing the most to total sales?

What we used: GROUP BY Product Name, SUM(Sales), ORDER BY DESC, LIMIT 10.

What the query does: Finds the ten biggest product-level revenue drivers.

04. REGIONAL PERFORMANCE

Business Question: Which regions are performing the best and worst in terms of sales and profitability?

What we used: GROUP BY Region, SUM(Sales), SUM(Profit), ORDER BY total_profit DESC.

What the query does: Compares geographical markets and identifies strong and weak regions.

05. MONTHLY SALES & PROFIT TREND

Business Question: How do sales and profit change across different months?

What we used: STR_TO_DATE(), MONTH(), SUM(), GROUP BY, ORDER BY.

What the query does: Converts the text date, extracts the month, and compares total sales and profit by month.

06. CUSTOMER SEGMENT PERFORMANCE

Business Question: Which customer segments contribute the most to overall sales and profit?

What we used: GROUP BY Segment, SUM(Sales), SUM(Profit), ORDER BY.

What the query does: Shows which customer groups are most valuable.

07. LOSS-MAKING PRODUCTS

Business Question: Which products are generating losses for the business?

What we used: GROUP BY Product Name, SUM(Profit), HAVING SUM(Profit) < 0, ORDER BY.

What the query does: Filters grouped products to identify those with negative total profit.

08. DISCOUNT & PROFITABILITY ANALYSIS

Business Question: How does the average discount affect profitability across categories?

What we used: AVG(Discount), SUM(Sales), SUM(Profit), GROUP BY Category, ORDER BY.

What the query does: Compares discount levels with sales and profit to evaluate discount strategy.

09. HIGH-VALUE CUSTOMERS

Business Question: Which customers have spent more than the average customer?

What we used: CTE, GROUP BY, SUM(), AVG(), subquery, WHERE, ORDER BY.

What the query does: Calculates each customer's spending, finds the average, then returns customers above that average.

10. YEAR-OVER-YEAR SALES GROWTH

Business Question: Is the company's sales performance improving or declining compared with the previous year?

What we used: CTE, YEAR(), STR_TO_DATE(), SUM(), LAG(), OVER().

What the query does: Compares each year's sales with the previous year to measure growth or decline.

WHAT THE RESULTS MEAN

Q1: Gives overall sales, profit, unique orders and quantity sold — the main business KPIs.

Q2: Shows which categories are strongest in revenue and profit.

Q3: Identifies the top 10 revenue-generating products.

Q4: Shows which regions are most and least profitable.

Q5: Shows monthly sales and profit patterns and possible seasonality.

Q6: Your result showed Consumer as the strongest segment, followed by Corporate and Home Office.

Q7: Identifies products with negative total profit so management can investigate pricing, discounts or costs.

Q8: Lets management compare average discount with sales and profit by category.

Q9: Identifies customers spending above the average customer, useful for retention and loyalty strategies.

Q10: Your result showed a decline in 2015 followed by strong sales growth in 2016 and 2017.

SQL SKILLS DEMONSTRATED

Basic: SELECT, SUM(), COUNT(), AVG(), GROUP BY, ORDER BY, LIMIT.

Intermediate: HAVING, date functions, STR_TO_DATE(), CTEs, subqueries.

Advanced: LAG() and window functions.

PRESENTATION INTRODUCTION

“My project is titled **Retail Revenue & Profit Intelligence**. The objective is to use **SQL** to analyze retail transaction data and convert it into useful business insights. I first performed data-quality checks and then created ten business-focused queries covering overall performance, categories, products, regions, monthly trends, customer segments, loss-making products, discount impact, high-value customers and year-over-year growth. The goal is not just to write SQL queries, but to use SQL to answer real business questions and support management decisions.”

PRESENTATION CLOSING

“Overall, this project demonstrates how SQL can move from raw retail data to actionable business intelligence. It helps identify revenue and profit drivers, loss areas, valuable customers, discount impact and business growth.”